Hercor College | Department of Computer Science | BSCS-2C
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PROPOSAL 1

CheynTech: Online Ordering and Delivery Management System for Cheyn's Gadgets

1. Executive Summary

Cheyn's Gadgets is a local gadget retailer that currently operates without an online sales channel and relies entirely on walk-in transactions and informal inventory tracking. This project proposes the development of a fully functional, live-deployed Online Ordering and Delivery Management System for Cheyn's Gadgets, a web-based platform that allows customers to browse and place orders online — for either in-store pickup or local delivery — while giving the store owner a real-time view of stock levels, orders, and transaction records. The system will be developed using HTML, CSS, JavaScript (front-end), PHP (back-end), and MySQL (database), and deployed on a public hosting platform with a registered domain. This directly addresses the competitive gap between Cheyn's Gadgets and digital-first competitors in the consumer electronics market.

2. Problem Statement

Cheyn's Gadgets has no online sales channel and no systemized inventory tracking, unlike competitors who have already shifted to e-commerce. The business currently depends on:

- **No digital product catalog:** customers can't browse, search, or view product details remotely and must visit the store in person just to see what's available. This creates friction for customers comparing options or checking stock, putting Cheyn's Gadgets at a disadvantage against competitors with online catalogs.
- **No remote purchasing:** even after finding a product, there's no way to complete a purchase without visiting the stores every sale still requires an in-person transaction.
- **No delivery option:** customers who can't visit the store in person have no way to receive a purchase, limiting sales to walk-in traffic only.
- **Untracked stockouts:** inventory is managed informally, leading to overselling and untracked product shortages.
- **No transaction records:** there is no digital log of purchases, making reconciliation and financial tracking difficult.

As consumers increasingly prefer online shopping, Cheyn's Gadgets risks losing market share to competitors with functional digital storefronts and automated inventory systems. There is a clear, verifiable operational problem that a web-based solution can directly solve.

3. Proposed Solution

We propose building a web-based Online Ordering and Delivery Management System for Cheyn's Gadgets with the following key features:

- **Customer-facing storefront:** A product catalog with categories, search/filter, and product detail pages, managed through an admin panel where the owner can add, edit, and delete products.
- **Shopping cart & checkout:** Add-to-cart functionality with order summary and checkout flow (manual payment confirmation, no automated gateway).
- **Order fulfillment (pickup or delivery):** At checkout, the customer chooses in-store pickup or local delivery. Delivery orders are assigned to staff and coordinated manually (no courier API or route-optimization integration); the "Completed" status reflects confirmed pickup or confirmed drop-off.
- **Real-time inventory tracking:** Automatic stock decrement on confirmed orders, with low-stock alerts and manual stock-adjustment controls in the admin panel.
- **Order management:** Transaction records with order history, status updates (Pending → Processing → Ready for Pickup / Out for Delivery → Completed), a customer-facing order status page, and an admin dashboard to view and update order statuses.

Target Audience: Existing and prospective customers of Cheyn's Gadgets (local consumers and walk-in customers looking for an online option), and the store owner/admin.

4. Technical Approach

Layer	Technologies
Front-End	HTML5, CSS3 (Bootstrap 5), JavaScript (Vanilla JS / Fetch API)
Back-End	PHP 8.x with PDO for database access
Database	MySQL — Products, Orders, Customers, Inventory tables
Security	HTTPS, bcrypt password hashing, PDO prepared statements (SQL injection prevention), XSS prevention, CSRF tokens
Hosting & Deployment	cPanel-based shared hosting, public domain registration
Design & Prototyping	Figma (wireframes → hi-fi prototype)
Version Control	Git / GitHub
Performance & SEO	Lighthouse audits (Performance ≥80 mobile, Accessibility ≥90), meta tags, sitemap.xml

5. Project Scope

In Scope

- Product catalog with categories, descriptions, images, and pricing
- Customer registration, login, and profile management
- Shopping cart and checkout (manual payment confirmation)
- Admin panel: product CRUD, stock management, order status updates
- Real-time stock level tracking with decrement on order confirmation
- Order status tracking (Pending / Processing / Ready for Pickup / Out for Delivery / Completed)
- Order fulfillment options: in-store pickup or staff-coordinated local delivery
- Live deployment with a registered domain
- Responsive design (tested at 375px, 768px, 1440px)
- SEO implementation (title tags, meta descriptions, sitemap, robots.txt)
- Security: HTTPS, bcrypt, prepared statements, XSS/CSRF protection

Out of Scope

- Automated payment gateway (e.g., PayMaya, GCash, Stripe)
- Automated courier/logistics integration (e.g., real-time GPS tracking, third-party delivery apps like Lalamove/Grab); delivery itself is in scope but staff-arranged and manually dispatched
- Multi-branch inventory support
- Mobile application (iOS/Android)
- Customer reviews and ratings system (Phase 2 feature)

6. Team Roles

Role	Responsibility
Marvin A. (Project M.)	Sprint planning, Gantt chart maintenance, milestone tracking, team coordination
Jayro G. (UI/UX D.)	Wireframes, hi-fi prototype (Figma), style guide, responsive layout design
Neo C. (Front end)	HTML/CSS/JS implementation, Bootstrap integration, Fetch API calls
Marvin A. (Scripting)	PHP scripting, PDO database integration, session management, authentication
Troy D. (Database	MySQL schema design, CRUD logic, data integrity, SQL export
Kim B. (Tester)	Testing across devices/browsers, Lighthouse audits, content population, SEO tags

Note: *In a small team, members will cross-fill roles. The Project Manager also contributes to QA; the Front-End Developer assists the UI/UX Designer.*

7. Timeline

Week	Milestone
Week 1–2	Project setup; proposal draft; finalize idea and scope
Week 3–4	Lo-fi wireframes; WBS and Gantt chart; IEEE research begins
Week 5–6	Hi-fi Figma prototype; style guide; IEEE Literature Review Report due
Week 7–8	Front-end development (HTML/CSS/JS); responsive layout; Prelim Exam
Week 9–10	PHP back-end setup; MySQL schema design; admin panel
Week 11–12	CRUD operations (products, orders, inventory); PDO prepared statements
Week 13–14	Security implementation; responsive testing; Midterm Exam / Midterm Build Due
Week 15–16	Performance optimization; Lighthouse audit; SEO implementation
Week 17–18	Deployment (domain + hosting); final testing; polish
Week 19	Project presentation deck preparation; peer evaluation
Week 20	Final Defense; Final Project Submission

8. Success Metrics

Metric	Target
Live Deployment	Publicly accessible URL with all features functional and realistic product data
Lighthouse Performance	≥80 on Mobile; ≥90 Accessibility

Metric	Target
Responsive Design	Fully functional at 375px (mobile), 768px (tablet), 1440px (desktop)
Security Compliance	HTTPS active; bcrypt passwords; no SQL injection vulnerabilities; CSRF tokens implemented
Admin Functionality	Owner can add/edit/delete products and update order status without developer intervention
Inventory Accuracy	Stock levels decrement correctly on confirmed orders; admin receives low-stock visibility
Feature Completion	All in-scope features pass QA checklist before final submission
Academic Requirements	Submitted with WBS, Gantt chart, IEEE Literature Review, presentation deck, and peer evaluation